Ryan Joshua George

Bangalore, India

ryang0304@gmail.com Ryan George github.com/ryang030403

Full Stack Developer with hands-on experience building scalable web applications and APIs; skilled in React.js, Node.js, and distributed systems, with an academic foundation in machine learning and data science.

Experience

Grassroots Jan 2026 – Present

Full Stack Developer *Bangalore, India*

- Engineer and ship full-stack features end-to-end across a React.js/Node.js stack, supporting an active production user base
- Design and build RESTful APIs powering business-critical workflows, working closely with the team to keep response times low
- Partner with product, design, and QA teams in an Agile workflow to ship multiple releases per sprint with zero critical post-release bugs
- Own database schema design and integration, applying indexing and query optimization to improve query performance
- Streamline deployment workflows using CI/CD pipelines, shortening release turnaround time and reducing manual deployment steps

Enhancio Aug 2024 – Jan 2025

Front-End Developer Intern *Bangalore, India*

- Built and shipped responsive UI components in React.js and JavaScript used across multiple production pages, contributing to faster page load times
- Partnered with senior developers and designers to redesign core UI/UX flows to improve usability and user engagement
- Diagnosed and resolved cross-browser compatibility issues, ensuring consistent rendering across Chrome, Firefox, and Safari
- Built and integrated reusable front-end components consumed across multiple internal projects, reducing duplicate code

Projects

Real-Time Chat Server | *Node.js, Socket.IO, Express, JWT*

- Built a real-time messaging server load-tested at 100 concurrent authenticated WebSocket connections: 1.9ms average and 3.3ms p95 broadcast latency (localhost)
- Implemented JWT authentication on REST and the WebSocket handshake, AES-256-GCM encryption of stored messages, and live cross-room notifications
- Built multi-device session handling, with smoke and load tests verifying 0 of 20,000 deliveries dropped under load

Resume Screening System | *Python, NLP, Scikit-learn*

- Built a TF-IDF candidate-ranking engine with keyword-gap reports, configurable thresholds and top-K shortlisting, backed by a 6-test pytest suite
- Built a held-out evaluation on 8,000 human-labeled resume/job pairs: 55.0% shortlisting accuracy vs a 48.7% baseline (ROC AUC 0.59), establishing a benchmark for embedding-based models

Microservices E-Commerce Platform | *Node.js, Express, Docker, JWT*

- Designed a distributed microservices architecture with 4 independently deployable services (user, product, payment, order) plus an API gateway, each owning its own data
- Built secure REST APIs with JWT authentication and Docker containerization for all five services, orchestrated via Docker Compose
- Load-tested the checkout saga at 50 concurrent orders: ~500 orders/sec, zero overselling, and compensating rollback verified across 25 concurrent declined payments

Advanced Chess Analytics Engine | *Python, Data Structures & Algorithms*

- Built a PGN analytics engine with python-chess: streaming parser, recursive move tree, and win/draw/loss breakdowns for 16 opening lines
- Trained a logistic-regression win-probability model evaluated on 793 real Lichess games with a game-level split: 61.6% held-out accuracy vs a 51% baseline (77% in endgames)

Education

Monash University Jan 2025 – Dec 2025

Master of Data Science: one year of coursework completed (degree not conferred) *Melbourne, Australia*

- Relevant Coursework: Machine Learning, Artificial Intelligence, Statistical Modeling, Data Analytics

Dayananda Sagar University 2021 – 2024

Bachelor of Computer Applications *Bangalore, India*

- Relevant Coursework: Data Structures, Algorithms, Databases, Computer Networks, Web Technologies

Technical Skills

Languages: JavaScript, TypeScript, Python, SQL, HTML/CSS
Frameworks & Libraries: React.js, Node.js, Express.js, REST APIs, WebSockets
Databases: MongoDB, MySQL, Oracle SQL, Redis
Tools & Platforms: Git, GitHub, Docker, Postman, VS Code, Linux, Firebase, CI/CD
Core Concepts: Data Structures & Algorithms, OOP, System Design, Microservices Architecture, Database Design, Machine Learning, NLP